Supplementary Information for *From soluble iodine to functional blockage in zinc–iodine flow-battery positive electrodes*

Reduced iodine-state framework, implementation map, molecular priors and mesoscale closures

Contents

S1 Governing equations of the continuum full-cell model

Roadmap. This section is the long-form specification of the continuum scaffold whose minimal closed set is stated in the main text (governing-balances subsection). We first map every modelled object to how it is produced (Table S1): which quantities are *solved* primary fields, which are algebraic *closures* of a solved field, and which are *prescribed* inputs. We then give the species, charge, kinetic and solid-phase balances (this section), the geometry and compression rule (Sec. S1.1), the electrolyte flow and permeability closure (Sec. S1.2), the areal basis of the faradaic source (Sec. S1.3), and the reservoir, boundary and initial conditions (Sec. S1.4). Each block expands the correspondingly-named relation of the main-text backbone; the SI symbols ε and $\varepsilon_{l,\mathrm{eff}} = \varepsilon - \varepsilon_{\mathrm{s}}$ are the main text's felt porosity ε_0 and running pore-liquid fraction ε_l. The main text uses S_{base} for the

Table S1: Implementation-to-theory map for the continuum full-cell model. Each object is classified by how it is produced in the native run, verified from the exported time series (baseline, $i_{\text{app}} = 400\,\text{A}\,\text{m}^{-2}$, $q : 0 \to 120\,\text{mAh}\,\text{cm}^{-2}$). "Solved" = primary state advanced by a balance/ODE; "consequence" = evaluated pointwise on a solved field; "closure" = algebraic function of a solved field; "prescribed" = imposed boundary input.

object	exported source	status	data evidence / wording
$c_R = \text{I}^-$	`cI_m_surf_avg`	solved (depleting)	$1688 \to 704\ \text{mol}\,\text{m}^{-3}$ (-58%); transported, consumed primary state
c_O ($\text{I}_3{}^-$, I_2Br^-, I_2(aq))	`cI2_surf_tot_avg`	solved	$104 \to 432\ \text{mol}\,\text{m}^{-3}$; $S : 0.19 \to 1.76$ crosses saturation
β (buffer)	`beta_surf_avg`	solved consequence	identity $\beta = 1 + K_{\text{I}} c_R + K_{\text{Br}} c_{\text{Br}}$ on solved c_R; $1263 \to 554$
ε_{s} (I_2(s))	`eps_s_avg`	solved (native)	$\partial_t \varepsilon_{\text{s}} = V_m(R_{\text{prec}} - R_{\text{diss}})$; $0 \to 3.2 \times 10^{-3}$
$\theta_{\text{eff}}, A_{\text{bare}}$	`theta_avg`	closure	single-valued in ε_{s}; $A_{\text{bare}} = 1 - \theta_{\text{eff}}$
$\varepsilon_{l,\text{eff}}, K, D, \kappa_l$	`eps_l_eff_avg`, `K_perm_rel_avg`	closure	algebraic in ε_{s} (Kozeny–Carman / Bruggeman); $K_{\text{rel}} : 1.00 \to 0.96$
$\mathbf{u}, p$ (flow)	`u_native_mag_avg`, `p_native_avg`	solved (two-way FPF)	feedback negligible: solved u varies $< 0.1\%$ as ε_{s} grows 3 decades; identical onset to a frozen-velocity control
cell voltage	`voltage_V`	solved (galvanostatic)	raw terminal $1.40 \to 1.73$ V; late rise emergent

within-electrolyte continuum state clock and S_{eff} for the composition-resolved risk coordinate; the chemistry-axis calibration of $\Lambda^{\text{rel}}_{\text{Br}}$ and $\chi^{\text{rel}}_{\text{ZnI}_2}$ is documented in Sec. S7 and is not part of the native COMSOL stress-field identity check.

The native full-cell model is a tertiary-current-distribution problem coupled to Darcy flow and a distributed solid-phase ordinary differential equation. With $R \equiv \text{I}^-$ and O the lumped oxidized-iodine pool ($\text{I}_3{}^-$, I_2Br^-, I_2(aq); bromide in excess so migration is folded into an effective formal potential), the porous-electrode species balances along the through-plane coordinate x are

$$\varepsilon\,\partial_t c_R = \nabla\cdot(D_{R,\text{eff}}\nabla c_R) - 2R_F, \tag{S1}$$

$$\varepsilon\,\partial_t c_O = \nabla\cdot(D_{O,\text{eff}}\nabla c_O) + R_F - \tfrac{1}{V_m}\,\partial_t\varepsilon_{\text{s}}, \tag{S2}$$

with $R_F = a_{\text{eff}} j_F/(2F)$ the volumetric faradaic rate (below). The net solid source $\frac{1}{V_m}\partial_t\varepsilon_{\text{s}} = R_{\text{prec}} - R_{\text{diss}}$ (solid ODE below) enters the c_O balance with the opposite sign, as the term $-\frac{1}{V_m}\partial_t\varepsilon_{\text{s}}$. Species storage uses the felt porosity ε in the dilute-solid approximation ($\varepsilon_{\text{s}} \ll \varepsilon$); the transport closures use $\varepsilon_{l,\text{eff}} = \varepsilon - \varepsilon_{\text{s}}$ (below). Charge is conserved in the solid and electrolyte phases, $\nabla\cdot\mathbf{i}_s = -a_{\text{eff}} j_F$, $\nabla\cdot\mathbf{i}_l = +a_{\text{eff}} j_F$, $\mathbf{i}_s = -\sigma_{s,\text{eff}}\nabla\phi_s$, $\mathbf{i}_l = -\kappa_{l,\text{eff}}\nabla\phi_l$, and concentration-dependent Butler–Volmer kinetics

$$j_F = k_{\text{BV}}\Big[c_R\,e^{(1-\alpha)nF\eta/RT} - c_O\,e^{-\alpha nF\eta/RT}\Big], \qquad \eta = \phi_s - \phi_l - E^{0\prime}, \tag{S3}$$

j_F being the local faradaic current density per unit *accessible* carbon area. The solid I_2(s) fraction evolves by a super-saturation source, $\partial_t\varepsilon_{\text{s}} = V_m\,(R_{\text{prec}} - R_{\text{diss}})$ with $R_{\text{prec}} \propto (S-1)_+$ and $R_{\text{diss}} \propto$

$(1-S)_+\,\varepsilon_{\rm s}$, and the algebraic feedback closures are

$$\varepsilon_{l,\rm eff} = \varepsilon - \varepsilon_{\rm s}, \quad K = \left(\tfrac{\varepsilon_{l,\rm eff}}{\varepsilon}\right)^3 \left(\tfrac{1-\varepsilon}{1-\varepsilon_{l,\rm eff}}\right)^2, \quad \{D, \kappa_l\} \propto \left(\tfrac{\varepsilon_{l,\rm eff}}{\varepsilon}\right)^{3/2}, \quad \theta_{\rm eff} = 1 - e^{-k_{\rm g} N^{1/3} \varepsilon_{\rm s}^{2/3}}. \tag{S4}$$

The reported COMSOL terminal voltage is a solved galvanostatic scaffold output, not a fitted polynomial (we verified that the empirical voltage-fit parameters present in the model file are not referenced by any solved variable); the experimental Stage-I/II voltage comparison instead uses the reduced observable layer of Sec. S4.

S1.1 Geometry, domains, and the compression rule

The native problem is posed in one through-plane dimension $x \in [0, L]$ spanning the compressed positive electrode of thickness L, bounded at $x = 0$ by the current collector and at $x = L$ by the ion-selective membrane, with a well-mixed tank reservoir (Sec. S1.4) feeding and draining the pore space. The deposited-solid fraction $\varepsilon_{\rm s}(x, t)$ is referred to the total electrode volume, consistent with the closure $\varepsilon_{l,\rm eff} = \varepsilon - \varepsilon_{\rm s}$ above (verified end-of-charge: $\varepsilon_{\rm s} + \varepsilon_{l,\rm eff} = 0.85$ to twelve figures). Mechanical compression is imposed at fixed fibre inventory, so squeezing the felt lowers L and ε together,

$$(1-\varepsilon)\,L = 0.30\,\text{mm} = \text{const}, \tag{S5}$$

and the compression series is a one-parameter family in L (equivalently ε) at constant solid loading.

S1.2 Electrolyte flow and the permeability closure

Convection in the pore space is a Darcy field layered on the solved electrode state,

$$\mathbf{u} = -\frac{\kappa_0\, K(\varepsilon_{\rm s})}{\mu}\,\nabla p, \qquad \nabla\cdot\mathbf{u} = 0, \tag{S6}$$

with $\mathbf{u}$ the superficial velocity, p the pore pressure, μ the viscosity, κ_0 the deposit-free permeability, and $K(\varepsilon_{\rm s})$ the relative-permeability closure of Eq. (S4) (Kozeny–Carman form, refined by the pore-network model of Sec. S3). The advective flux $\mathbf{u}\,c_i$ enters the species balances above; under excess bromide the migration of the charged iodine species is folded into $E^{0\prime}$, so the transport operator carries diffusion and advection only. The velocity is solved *together with* the electrochemistry: the native porous-media-flow interface solves $\mathbf{u}, p$ and feeds the solved $\mathbf{u}$ back into the species convection (full two-way coupling), with the permeability multiplier K/K_0 a smooth, monotone, single-valued closure of the solved solid inventory (unity at $\varepsilon_{\rm s} = 0$, decreasing thereafter). In practice this feedback is negligible: across a full charge the solved $\mathbf{u}$ is constant to $<0.1\%$ even as $\varepsilon_{\rm s}$ grows by three orders of magnitude, and a control solve with the velocity frozen gives the *identical* failure onset. The reason is that K/K_0 stays ≈ 1 until the accessibility threshold is already crossed ($K/K_0 = 0.998$ at $\theta_{\rm eff} = \frac{1}{2}$), so within the analysed pre-failure window the pressure rises by only a few percent. Far past failure the feedback is no longer benign: as the pore network fully clogs ($K/K_0 \to 0$, reaching ≈ 0.25 in the most diffusion-limited case at $\theta_{\rm eff} \to 1$) the Darcy pressure diverges super-linearly, rising $\sim 10^3\times$ toward ~ 8 bar in the frozen-throughput limit — the regime in which constant-flow pumping mechanically ruptures tubing and seals in practice. This extrapolation is reported only as a post-failure hydraulic consequence, not as a fitted or validated operating threshold. That fully-destabilised hydraulic runaway is a distinct, deep-post-failure mode downstream of the accessibility-collapse failure analysed here and lies outside the present continuum scope; we therefore use the in-window solution to quantify the (mild) hydraulic consequence of I_2(s) accumulation, and make no claim of a flow-coupled voltage validation.

S1.3 Areal basis of the faradaic source and the through-thickness reduction

The faradaic reaction is distributed over the locally accessible interfacial area. With a_s the geometric specific area of the bare matrix and θ_{eff} the accessibility-loss closure, the active-area density and the volumetric oxidized-iodine generation are

$$a_{\mathrm{eff}} = a_s\,(1-\theta_{\mathrm{eff}}), \qquad R_{\mathrm{gen}} = R_F = \frac{a_{\mathrm{eff}}\, j_F}{2F}, \tag{S7}$$

with j_F the local faradaic current density per unit accessible area, so the volumetric faradaic *current* is $a_{\mathrm{eff}} j_F = 2FR_F$ and the same R_F appears in the species balances above. Integrating across the electrode thickness under galvanostatic control,

$$J_{\mathrm{gen}} = \int_0^L R_{\mathrm{gen}}\,\mathrm{d}x = \frac{1}{2F}\int_0^L a_{\mathrm{eff}}\, j_F\,\mathrm{d}x = \frac{i_{\mathrm{app}}}{2F}, \tag{S8}$$

the last equality being charge balance: the through-thickness integral of the distributed faradaic current equals the imposed areal current density i_{app}, independent of the x-profile of a_{eff}. This is why the reduced generation group Π_{gen} of the main text is built from $i_{\mathrm{app}}/2F$ and carries no explicit $a_s L$ factor: the specific area enters the local rate Eq. (S7) but cancels from the integrated areal supply Eq. (S8). Accessibility loss reshapes *where* iodine is generated, not the total areal generation rate, which is pinned to the applied current.

S1.4 Reservoir coupling, boundary, and initial conditions

The pore electrolyte is coupled to a well-mixed tank of volume V_{tank} at concentrations $c_i^{\mathrm{tank}}(t)$,

$$V_{\mathrm{tank}}\,\frac{\mathrm{d}c_i^{\mathrm{tank}}}{\mathrm{d}t} = \dot{V}\big(c_i^{\mathrm{out}} - c_i^{\mathrm{tank}}\big), \tag{S9}$$

with $\dot{V}$ the volumetric flow rate and c_i^{out} the electrode outlet concentration. The boundary conditions are

$$\text{inlet:}\quad c_i = c_i^{\mathrm{tank}}, \quad \mathbf{u}\cdot\hat{\mathbf{n}} = u_{\mathrm{in}}; \tag{S10}$$

$$\text{outlet:}\quad -D_{i,\mathrm{eff}}\nabla c_i\cdot\hat{\mathbf{n}} = 0, \quad p = p_{\mathrm{out}}; \tag{S11}$$

$$\text{collector } (x{=}0):\quad \mathbf{i}_s\cdot\hat{\mathbf{n}} = i_{\mathrm{app}}, \quad \mathbf{i}_l\cdot\hat{\mathbf{n}} = 0, \quad \nabla c_i\cdot\hat{\mathbf{n}} = 0; \tag{S12}$$

$$\text{membrane } (x{=}L):\quad \mathbf{i}_s\cdot\hat{\mathbf{n}} = 0, \quad \mathbf{i}_l\cdot\hat{\mathbf{n}} = i_{\mathrm{app}}, \quad \nabla c_{R,O}\cdot\hat{\mathbf{n}} = 0, \tag{S13}$$

with $\hat{\mathbf{n}}$ the outward normal: the iodine species are blocked at the membrane (only the charge-balancing counter-ion crosses, carried by i_{app}), electronic current is collected at $x = 0$, and ionic current closes the circuit at $x = L$. The deposited I_2(s) obeys a purely local (no-flux) condition, consistent with its distributed ODE $\partial_t\varepsilon_{\mathrm{s}} = V_m(R_{\mathrm{prec}} - R_{\mathrm{diss}})$. The initial state is start-of-charge equilibrium: uniform $c_R(x,0) = c_{R,0}$, $c_O(x,0) = c_{O,0}$, zero solid $\varepsilon_{\mathrm{s}}(x,0) = 0$ (hence $\theta_{\mathrm{eff}} = 0$, $a_{\mathrm{eff}} = a_s$, $K = 1$), and the rest potential consistent with $c_{R,0}, c_{O,0}$. In the solved baseline the surface iodide c_R depletes monotonically over charge ($1688\,\mathrm{mol\,m^{-3}}\rightarrow 704\,\mathrm{mol\,m^{-3}}$, -58%) as the oxidized pool grows and I_2(s) accumulates, so c_R is a transported, *depleting dependent variable*, and the saturation-dependent quantities that follow from it—including the buffer β—are solved consequences of the species, charge, kinetic, and solid-phase relations above, not reconstructed inputs. This final manuscript branch — the reduced two-species (c_R/c_O) scaffold — therefore supersedes any earlier reduced treatment that held I^- fixed (the corrected-covfix constant-I^- export): here I^- is a solved, depleting primary field (Table S1), and every β-, S_{base}-, and R^2=1.000 statement in the main text is evaluated on that solved field, not on a reconstructed buffer.

S2 Molecular priors

Table S2: Periodic DFT (D3+PBE) I_2 adsorption and coalescence energies on graphitic slabs. Used as trapping/coalescence *tendencies* (nucleation placement bias), never as kinetic constants.

site	E_{ads} (eV)	note
basal pristine	−0.40	baseline
carbonyl C=O	−0.85	strong O trap
hydroxyl C−OH	−1.00	strongest accepted trap
single vacancy	−0.32	weaker than basal
$2I_2$ coalescence at C−OH	−0.69	coalescence favored

Table S3: Molecular-dynamics carrier diffusivities in 1 M ZnI_2 + 4 M NH_4Br ($\times 10^{-9}$ m^2 s^{-1}). A bounded prior; only the relative ordering is used.

species	q=0.8 ECC	q=1.0
H_2O	1.99	1.33
I^-	1.22	0.71
I_2	1.07	0.59
$I_3{}^-$	0.75	0.53
I_2Br^-	0.72	0.52

S3 Mesoscale closures

The accessibility closure $\theta_{\mathrm{eff}}(\varepsilon_{\mathrm{s}})$ and the relative permeability $K(\varepsilon_{\mathrm{s}})$ (Fig. S3) are obtained from the 3D fiber and pore-network models. At the operating solid fraction $\varepsilon_{\mathrm{s}} \approx 3 \times 10^{-3}$ the network permeability stays within a few percent of unity for all deposition laws, confirming that hydraulic clogging is not the failure mode. The dense-nucleation fiber closure $\theta_{\mathrm{eff}} = 1 - \exp(-35.4\,\varepsilon_{\mathrm{s}}^{0.622})$ (exponent $\approx 2/3$, consistent with Eq. (2) of the main text) crosses half-coverage at $\varepsilon_{\mathrm{s}} = 1.8 \times 10^{-3}$; the independent spatial-percolation threshold $\varepsilon_{\mathrm{s}}^{*} = 2.2 \times 10^{-3}$ (dense nucleation, 10^{14} m^{-2}) agrees with this to within the nucleation-density uncertainty, and both are the functional-blocking fraction used in the main text. The lower main-text θ_{eff} values (0.11–0.42 over $\varepsilon_{\mathrm{s}} = 3.4 \times 10^{-4}$–$3.4 \times 10^{-3}$) are the diffusion-resolved 3D-fibre accessibility, which lies below this algebraic envelope.

S4 Voltage: coverage–Tafel degeneracy and internal overpotential

The absolute charge voltage does not uniquely determine the coverage morphology (Fig. S4): the accessibility contribution to the real pre-failure rise ($\sim$ 61 mV) is reproduced by end-of-charge coverage $\theta_{\mathrm{eff}} \in [0.6, 0.9]$ with a physical Tafel slope, while a near-complete film would require an unphysically small slope. The internal overpotential decomposition (Fig. S5) shows the late voltage response is carried by the electronic/contact and accessibility terms, with the ionic term small and the relative permeability near unity.

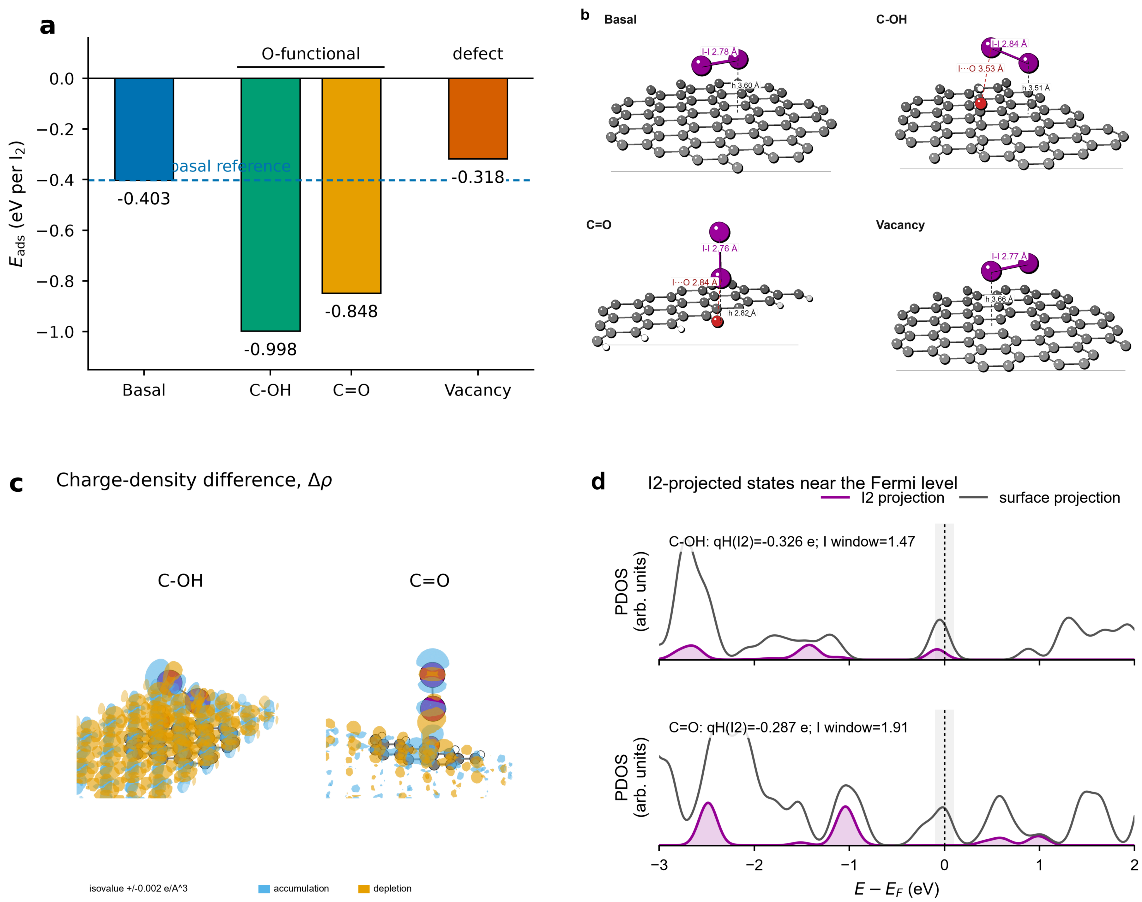

Figure S1: Periodic CP2K single-I_2 DFT. (a) Adsorption energy per I_2 across basal, hydroxyl, carbonyl and vacancy sites; (b) optimized adsorption geometries; (c) charge-density-difference $\Delta\rho$ (accumulation blue, depletion orange) at C−OH and C=O; (d) I_2-projected density of states near the Fermi level. The maps confirm physisorption-dominated, oxygen-site-enhanced I_2 retention, used only as a bounded placement prior.

S5 Experimental configuration and data provenance

All experiments used only two reagents in aqueous solution, ZnI_2 and NH_4Br; no other electrolyte chemistry was employed. Two cell formats appear. (i) *Full cells* (Neware galvanostatic cycling) use a $4\,cm^2$ positive iodine felt against a Zn negative electrode with a Nafion 212 separator; the areal capacities quoted in the main text and in the compression table below are *max-deliverable* discharge values read directly from the raw `.ndax` records (parsed independently), not a coulombic-efficiency-binned cut-off. (ii) *Three-electrode tests* (cyclic voltammetry, chronoamperometry, impedance) use the positive iodine felt as the $1\,cm^2$ working electrode with an Ag/AgCl reference electrode; they enter only as operating-regime and electrode-resolved observable anchors, never as fitted inputs.

Instrument-metadata disclaimer. The electrochemical-workstation project files (`.mps`) carry a *predecessor template* in their stored metadata — a saturated-calomel reference, an Au/Pd electrode, a NaCl + ferri/ferrocyanide electrolyte, and a $0.001\,cm^2$ area — that does *not* describe the present experiments. That template is stale and is to be ignored; the operative configuration is the three-

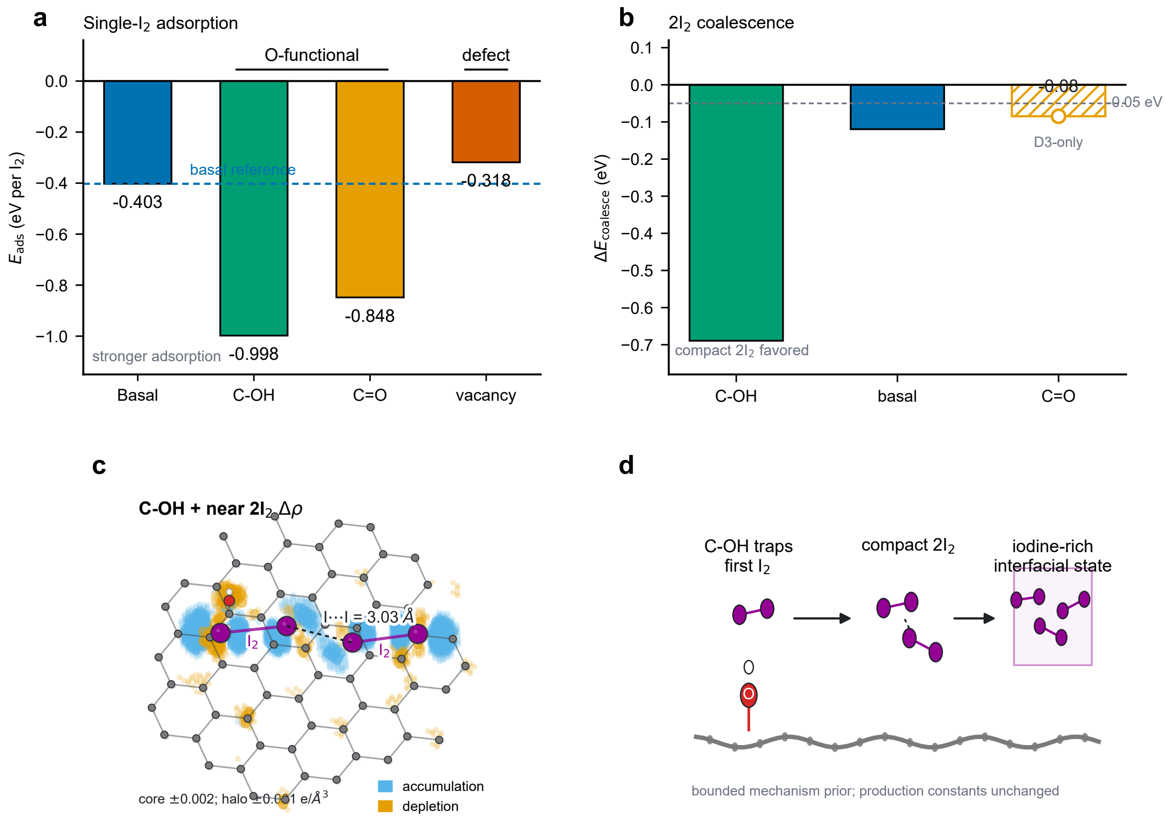

Figure S2: Periodic CP2K $2I_2$ coalescence. (a) Single-I_2 adsorption; (b) $2I_2$ coalescence energy (compact pair favoured at C−OH, −0.69 eV); (c) charge-density-difference $\Delta\rho$ for the C−OH-bound I_2 pair (I–I = 3.03 Å); (d) mechanism: oxygenated sites trap the first I_2, seed a compact $2I_2$ cluster and an iodine-rich interfacial state—a bounded mechanism prior, with production constants unchanged.

electrode Ag/AgCl, 1 cm^2 iodine-felt working electrode above (full cell 4 cm^2, ZnI_2+NH_4Br only). All reported capacities and electrode-resolved trends use the operative configuration; the impedance attempts noted in Sec. S9 are likewise on this configuration.

S6 Experimental compression series

S7 Chemistry-axis fit and identifiability

The cross-composition capacity model of the main text,

$$Q_{\mathrm{f}} = \min\bigl(Q_{\mathrm{cap}},\, A/(S_{\mathrm{eff}}-1)_{+}\bigr), \tag{S14}$$

$$S_{\mathrm{eff}} = \frac{\kappa\, i_{\mathrm{app}}\,(1+\chi_I^{\mathrm{app}} c_{\mathrm{I}})}{c_{\mathrm{sat}}\,(1+K^{\mathrm{app}}_{\mathrm{sol,Br}} c_{\mathrm{Br}})}, \tag{S15}$$

is fitted to the eight matched-protocol full-cell points (five-point NH_4Br axis at 20 mA cm^{-2}, 1 M ZnI_2; three-point ZnI_2 axis at 40 mA cm^{-2}, 4 M NH_4Br). The joint fit gives $R^2 = 0.998$, against $R^2 = 0.054$ for a β-only null that moreover predicts the *wrong* bromide trend (0.97× over 0–4 M versus the observed 2.53×). A bootstrap with leave-one-out cross-checks (Table S5) sets what may be claimed:

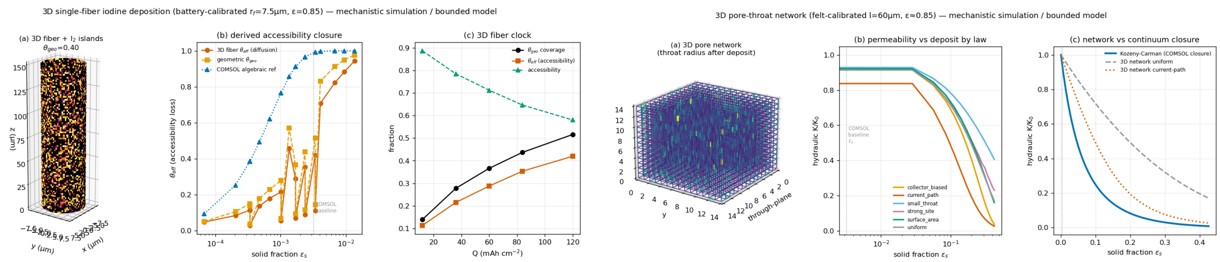

Figure S3: Mesoscale closures. Left: 3D single-fiber accessibility loss $\theta_{\rm eff}(\varepsilon_{\rm s})$ versus nucleation morphology. Right: 3D pore-network relative permeability versus retained fraction, compared to the continuum Kozeny–Carman law; both remain near unity at the operating point.

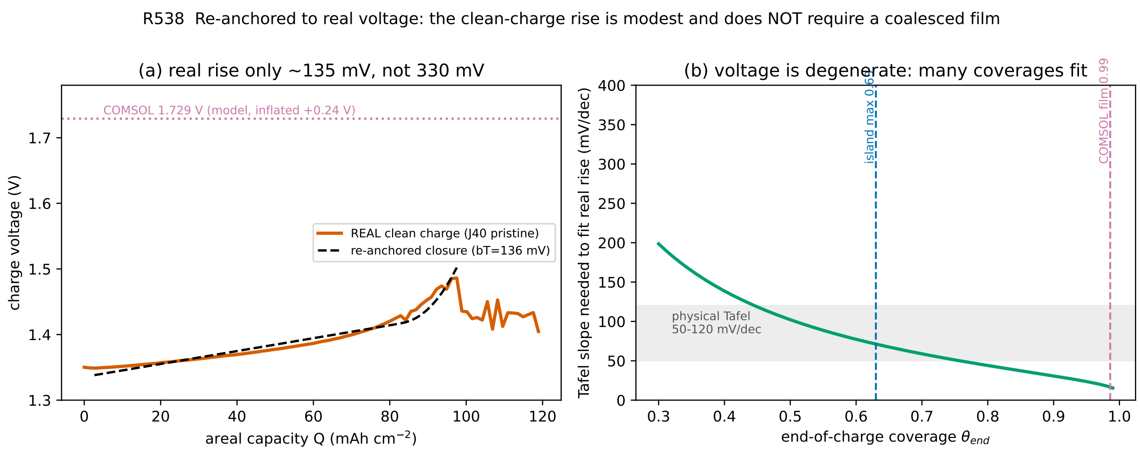

Figure S4: Coverage–Tafel degeneracy. Left: the real clean-charge rise is modest ($\sim$135 mV). Right: the accessibility contribution is reproduced over a wide coverage range by trading the Tafel slope; a coalesced film is not required by the voltage.

the intrinsic ceiling $Q_{\rm cap}$ and the bromide-accommodation coefficient $K^{\rm app}_{\rm sol,Br}$ are robustly identified (CI width/value < 0.4, stable across capacity definitions), whereas the precipitation strength A, prefactor κ, and iodide coefficient $\chi^{\rm app}_I$ are sloppy. The iodide axis fits three active parameters to three points and is confounded by the large ionic-strength change from 1 to $3\,\mathrm{M}\,\mathrm{ZnI_2}$; we therefore report the bromide-accommodation leg quantitatively and the iodide leg as a directional trend only.

S8 Key parameters

S9 Limitations and scope

The continuum model is a mechanism scaffold, not an exact voltage validator: its absolute charge voltage exceeds the real clean-charge ceiling by $\sim$0.24 V because the empirical accessibility closure over-blocks at high capacity; we therefore use it only for mechanism ordering and re-anchor all quantitative voltages to the data. The mesoscale closures are bounded geometric models, not imaged microstructures. The molecular priors set tendencies, not constants. The compression scaling and the rate threshold rest on a limited series ($N = 4$) and on an independent-laboratory rate constant in a different electrolyte, so they are order-of-magnitude consistency tests, not fits. We attempted impedance spectroscopy in the failure regime and obtained non-interpretable spectra, consistent

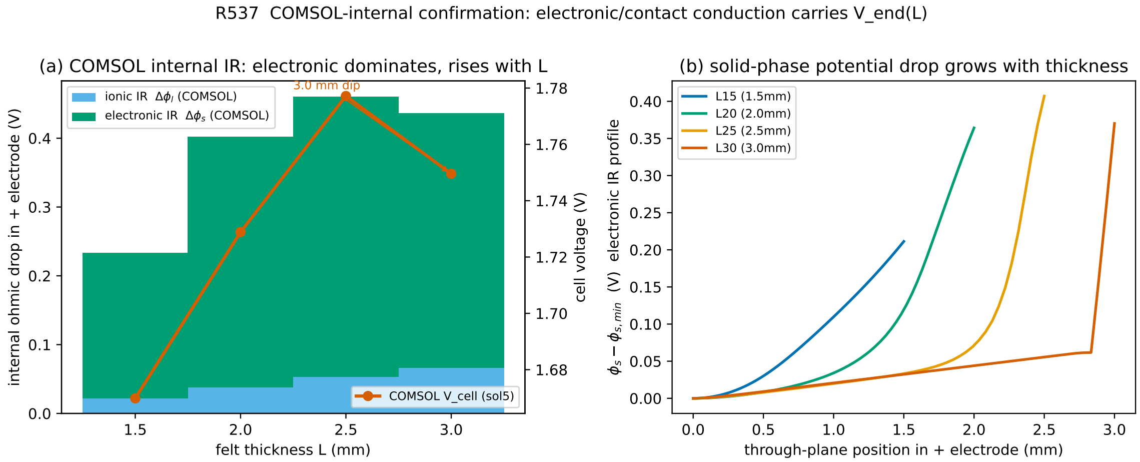

Figure S5: Internal overpotential decomposition of the continuum model across the compression series: the electronic/contact term dominates the thickness trend; the ionic ohmic term is small.

Table S4: Curated capacity-limit cycling of the compression series at $\sim 40\,\mathrm{mA\,cm^{-2}}$, $4\,\mathrm{cm^2}$, $1\,\mathrm{M}\,ZnI_2$ + $4\,\mathrm{M}\,NH_4Br$, Nafion 212. Areal capacities are discharge values; $\varepsilon L = L - 0.30$ mm is the pore volume per area.

L (mm)	ε	εL (mm)	max cap ($\mathrm{mAh\,cm^{-2}}$)	CE (%)	gap (V)
1.5	0.800	1.20	62.2	77.7	0.26
2.0	0.850	1.70	98.5	82.1	0.19
2.5	0.880	2.20	96.3	96.3	0.23
3.0	0.900	2.70	136.6	97.6	0.21

with the unstable, oscillating dissolution-limited state reported independently [1]; we therefore do not claim a failure-regime impedance signature.

References

[1] Yunhe Zhao, Yang Li, Jiatao Mao, Zhibin Yi, Nauman Mubarak, Yiting Zheng, Jang-Kyo Kim, and Qing Chen. Accelerating the dissolution kinetics of iodine with a cosolvent for a high-current zinc–iodine flow battery. *Journal of Materials Chemistry A*, 10(26):14090–14097, 2022. doi: 10.1039/d2ta03195g.

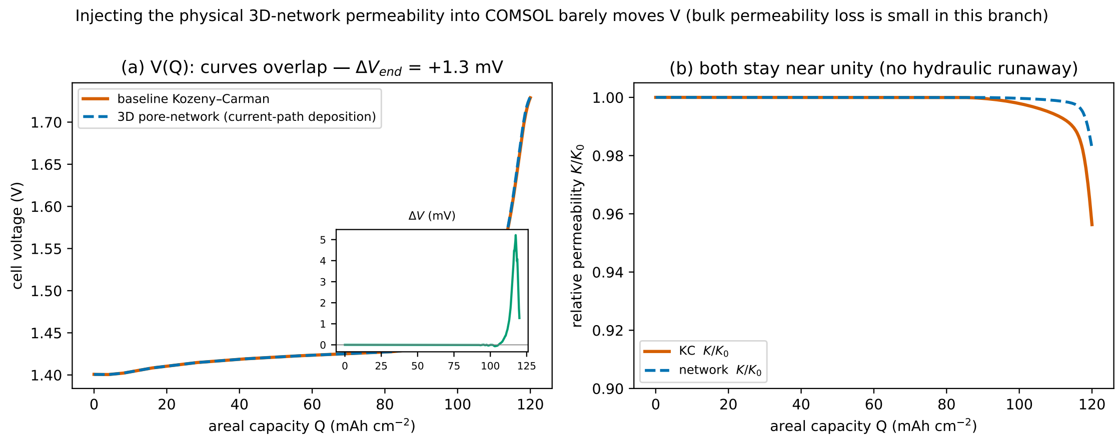

Figure S6: Bulk permeability loss is small within the smooth network closure. Replacing the mean-field Kozeny–Carman permeability with the physically derived pore-network closure $K(\varepsilon_s)$ and re-solving the continuum model shifts the cell voltage by only $\sim 1.3\,\mathrm{mV}$; the relative permeability stays near unity throughout (Kozeny–Carman endpoint 0.96, pore-network endpoint 0.98). This rules out a smooth, mean-field permeability penalty as the Stage-I/II driver, but does not exclude discrete or local pore-throat blockage.

Table S5: Bootstrap identifiability of the chemistry-axis model. A 95% confidence-interval width below 0.4× the central value is taken as robustly identified. $K^{\mathrm{app}}_{\mathrm{sol,Br}}$ is an apparent capacity-level iodine-accommodation coefficient (lumping solubility, $\mathrm{I_2Br^-}$-mediated dissolution, activity, and deposit morphology/reversibility), *not* the microscopic $\mathrm{I_2 + Br^- \rightleftharpoons I_2Br^-}$ association constant ($\sim 10\,\mathrm{M^{-1}}$).

parameter	value	95% CI	width/value (verdict)
Q_{cap} ($\mathrm{mAh\,cm^{-2}}$)	136.9	$[132.6, 140.6]$	0.06 (robust)
$K^{\mathrm{app}}_{\mathrm{sol,Br}}$ ($\mathrm{M^{-1}}$)	0.52	$[0.46, 0.60]$	0.27 (robust)
χ_I^{app} ($\mathrm{M^{-1}}$)	0.37	$[0.23, 0.66]$	1.14 (weak / directional)
κ	0.19	$[0.14, 0.50]$	1.84 (sloppy)
A	219	$[115, 900]$	3.58 (sloppy / degenerate)

Table S6: Central physical parameters.

quantity	symbol	value
electrode area	A	$4\,\mathrm{cm^2}$
porosity rule	$\varepsilon(L)$	$1 - 0.30\ \mathrm{mm}/L$
specific area	a_s	$4 \times 10^4\ \mathrm{m^{-1}}$
charge current density	i_{app}	$400\,\mathrm{A\,m^{-2}}$
$\mathrm{I_2}$ saturation	c_{sat}	$1.33\,\mathrm{mol\,m^{-3}}$
$\mathrm{I_2}$ molar volume	V_m	$5.15 \times 10^{-5}\ \mathrm{m^3\,mol^{-1}}$
speciation constants	$K_{\mathrm{I}}, K_{\mathrm{Br}}$	$0.72,\ 0.0115\ \mathrm{m^3\,mol^{-1}}$
Nernst layer	δ	$25\,\mathrm{\mu m}$
dissolution rate const. [1]	k	1.2–$2.2 \times 10^{-6}\ \mathrm{mol\,cm^{-2}\,s^{-1}}$